

## Computer Science &gt; Machine Learning

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## Neural Discrete Representation Learning

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Learning useful representations without supervision remains a key challenge in machine learning. In this paper, we propose a simple yet powerful generative model that learns such discrete representations. Our model, the Vector Quantised-Variational AutoEncoder (VQ-VAE), differs from VAEs in two key ways: the encoder network outputs discrete, rather than continuous, codes; and the prior is learnt rather than static. In order to learn a discrete latent representation, we incorporate ideas from vector quantisation (VQ). Using the VQ method allows the model to circumvent issues of "posterior collapse" – where the latents are ignored when they are paired with a powerful autoregressive decoder – typically observed in the VAE framework. Pairing these representations with an autoregressive prior, the model can generate high quality images, videos, and speech as well as doing high quality speaker conversion and unsupervised learning of phonemes, providing further evidence of the utility of the learnt representations.

Subjects: **Machine Learning (cs.LG)**

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## Neural Discrete Representation Learning

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# Introduction

## Background and Motivation

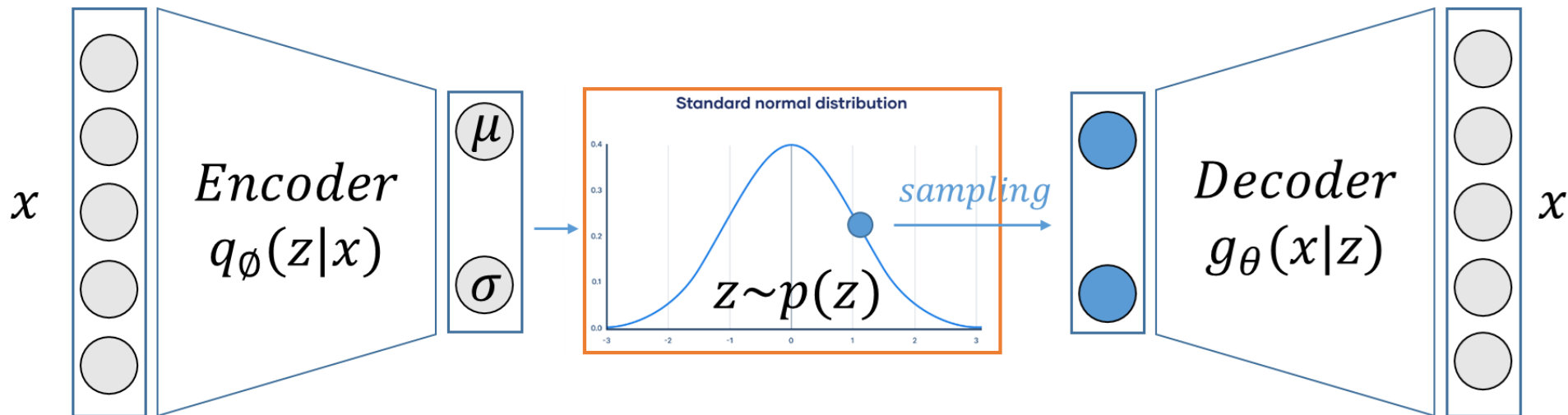
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- Learning useful representations without supervision remains a key challenge in machine learning. Learning representations with continuous features have been the focus of many previous work.
- However we concentrate on discrete representations which are potentially a more natural fit for many of the modalities we are interested in.
- Introducing the VQ-VAE model, which is simple, uses discrete latents, does not suffer from “posterior collapse” and has no variance issues.
- We show that a VQ-VAE perform as well as its continuous model counterparts in log-likelihood.
- When paired with a powerful prior, our samples are coherent and high quality on a wide variety of applications such as speech and video generation.
- We show evidence of learning language through raw speech, without any supervision, and show applications of unsupervised speaker conversion.

# Related work

## VAE

$$\underbrace{P(\theta|X)}_{\text{A posterior probability}} = \frac{\overbrace{P(X|\theta)}^{\text{Likelihood}} \overbrace{p(\theta)}^{\text{A prior probability}}}{\underbrace{P(X)}_{\text{Marginal probability}}}$$



- discrete VAEs - NIVL, VIMCO
- a new continuous reparametrisation - Concrete or Gumbel-softmax distribution.
- Autoregressive decoder - PixelCNNs
- Image compression
- most of these techniques are typically evaluated on relatively small datasets such as MNIST, and the dimensionality of the latent distributions is small (e.g., below 8).
- In our work, we use three complex image datasets (CIFAR10, ImageNet, and DeepMind Lab) and a raw speech dataset (VCTK).

# Part 3, VQ-VAE Architecture

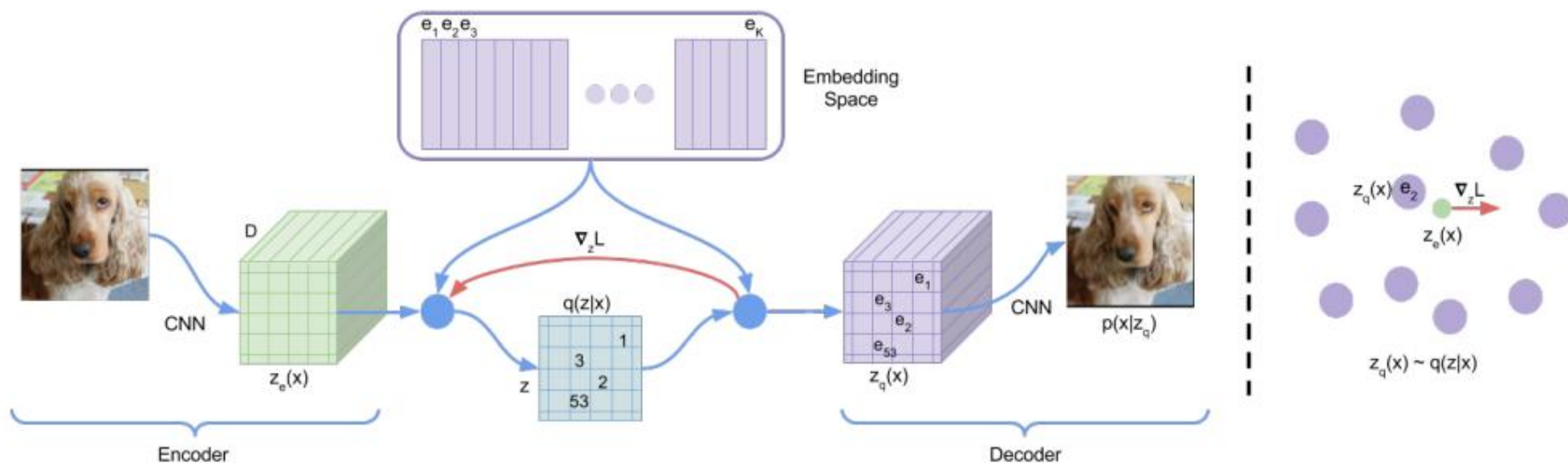


Figure 1: Left: A figure describing the VQ-VAE. Right: Visualisation of the embedding space. The output of the encoder  $z(x)$  is mapped to the nearest point  $e_2$ . The gradient  $\nabla_z L$  (in red) will push the encoder to change its output, which could alter the configuration in the next forward pass.

## Discrete Latent variables

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$$q(z = k|x) = \begin{cases} 1 & \text{for } k = \operatorname{argmin}_j \|z_e(x) - e_j\|_2, \\ 0 & \text{otherwise} \end{cases}, \quad (1)$$

$$z_q(x) = e_k, \quad \text{where } k = \operatorname{argmin}_j \|z_e(x) - e_j\|_2 \quad (2)$$

- latent embedding space  $e \in \mathbb{R}^{K \times D}$ 
  - where K is size of discrete latent space
  - D is dimensionality of each latent embedding vector  $e_i$
- D is dimensionality of each latent embedding vector

# VQ-VAE

## Learning

$$q(z = k|x) = \begin{cases} 1 & \text{for } k = \operatorname{argmin}_j \|z_e(x) - e_j\|_2, \\ 0 & \text{otherwise} \end{cases}, \quad (1)$$

$$z_q(x) = e_k, \quad \text{where } k = \operatorname{argmin}_j \|z_e(x) - e_j\|_2 \quad (2)$$

$$L = \boxed{\log p(x|z_q(x))} + \boxed{\|\operatorname{sg}[z_e(x)] - e\|_2^2} + \boxed{\beta \|z_e(x) - \operatorname{sg}[e]\|_2^2}, \quad (3)$$

- There is no real gradient defined for EQ 2.
- However we approximate the gradient similar to the straight-through estimator and just copy gradients from decoder input  $z_q(x)$  to encoder output  $z_e(x)$ .
- Forward - the nearest embedding  $z_q(x)$  is passed to the decoder.
- Backward - the gradient  $\nabla_z L$  is passed unaltered to the encoder.
- EQ3.
  - 1<sup>st</sup> term – reconstruction loss
  - 2<sup>nd</sup> term –  $l_2$  loss
  - 3<sup>rd</sup> term - commitment loss

# Experiments

## Images



Figure 2: Left: ImageNet 128x128x3 images, right: reconstructions from a VQ-VAE with a 32x32x1 latent space, with K=512.



Figure 3: Samples (128x128) from a VQ-VAE with a PixelCNN prior trained on ImageNet images. From left to right: kit fox, gray whale, brown bear, admiral (butterfly), coral reef, alp, microwave, pickup.

Samples drawn from the PixelCNN were mapped to pixel-space with the decoder of the VQ-VAE and can be seen in Figure 3.



Figure 4: Samples (128x128) from a VQ-VAE with a PixelCNN prior trained on frames captured from DeepMind Lab.

# Experiments

## Audio

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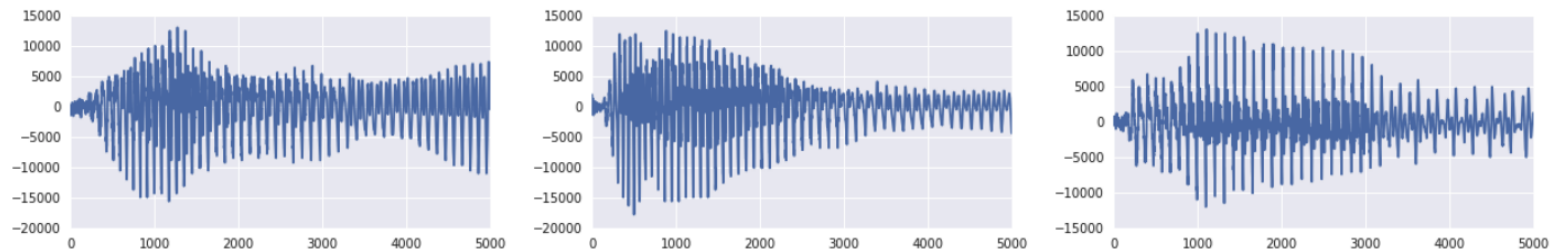


Figure 6: Left: original waveform, middle: reconstructed with same speaker-id, right: reconstructed with different speaker-id. The contents of the three waveforms are the same.

## Part 4, Experiments

### Video

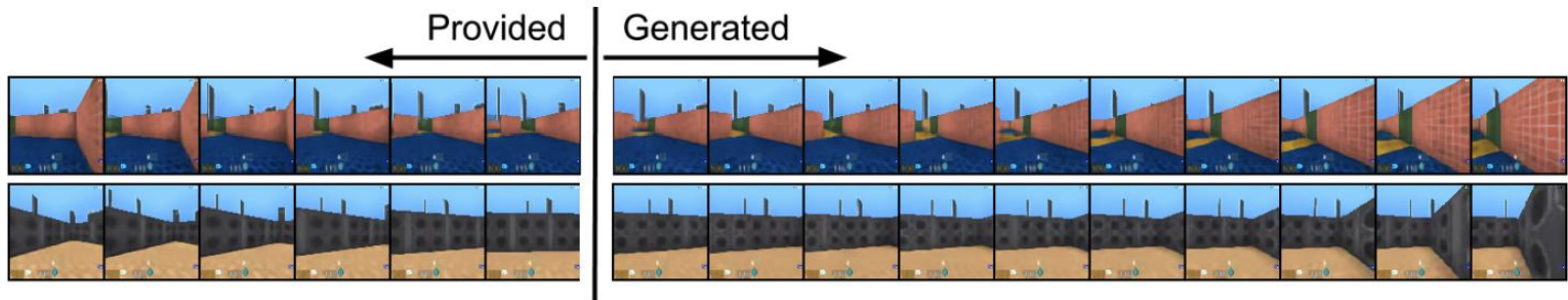


Figure 7: First 6 frames are provided to the model, following frames are generated conditioned on an action. Top: repeated action "move forward", bottom: repeated action "move right".

- **VQ-VAE Introduction:** Proposed a novel model that combines Variational Autoencoders (VAEs) with vector quantisation to produce discrete latent representations.
- **Long-Term Dependency Modeling:** Demonstrated the ability of VQ-VAEs to model long-range dependencies across various domains:
  - Generation of  $128 \times 128$  color images.
  - Action-conditional video prediction.
  - Unconditional speech generation and speaker conversion.
- **Unsupervised Feature Learning:** Showed that VQ-VAEs learn meaningful, high-level features in a fully unsupervised manner.
- **Competitive Performance:** Achieved likelihoods comparable to continuous latent variable models on the CIFAR-10 dataset.
- **Speech Representation:** First model to learn discrete speech representations related to phonemes without supervision.